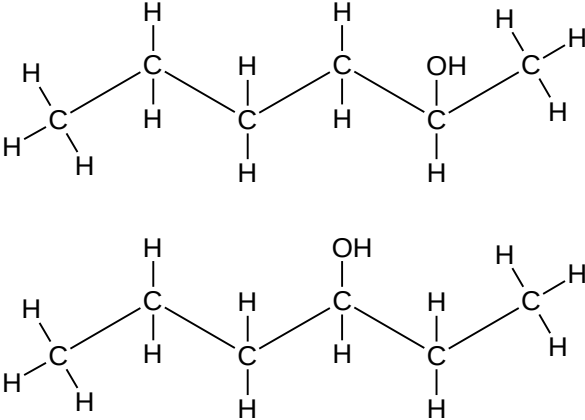
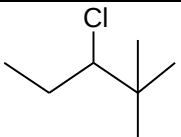
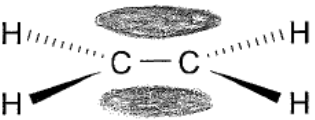
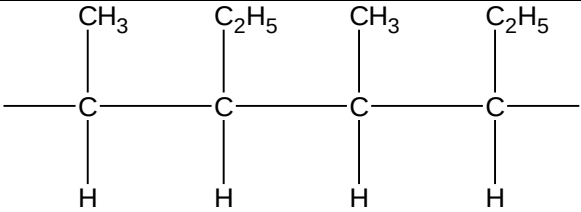
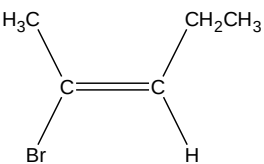
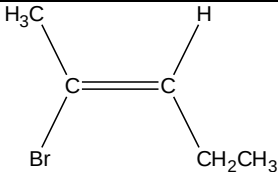


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				any secondary alcohol containing six carbon atoms e.g. 		1		1		
2.						1		1		
3.				only ethanol forms hydrogen bonds / ethane does not form hydrogen bonds but ethanol does (1) with water (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.				π orbital above and below plane of molecule 	1			1		
5.	(a)			$2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$ ignore state symbols	1			1		
	(b)			yeast / zymase	1			1		1
6.						1		1		
7.				 E  Z correct formulae (1) E and Z labelled correctly (1) award (1) for one correct formula with correct label		2		2		
Section A total					5	5	0	10	0	1